

**PART1**

Choose the answer of the following question's and transfer it in CAPITAL letters to table

Q1: If You have left over some unused chemicals after you have completed the lab, you should:

- A- Place liquids and solids in the trash always.
- B- Put left over chemicals back into their original containers. ✗
- C- Pour all the chemicals in the sink
- ☒ D- Follow all instructions given for disposal of chemicals.

Q2: According to solubility rules, which of the following statements is correct:

- A- Adding large amount of solute will show a precipitate because the solution reaches saturation.
- ☒ B- All group 1 salts are water insoluble. ✓
- C- Not all Nitrate (NO_3^-) salts are water soluble. ✗
- D- Increasing temperature of solvent will decrease the solubility of solid solute.

Q3: A suitable equipment for measuring volume of solid in density experiment is:

- A- Volumetric flask
- B- Pycnometer
- ☒ C- Balance
- D- Graduated cylinder

Q4: Measuring the boiling point of a compound can be used to _____

- A- Confirm the compound's formula ✗
- ☒ B- Check the compound's purity
- C- Determine the compound's structure ✗
- D- Calculate the compound's density ✗

Q5: Suppose a boiling point experiment for a liquid is performed at the Dead sea, and on the top of Rum mountains. which of the following statements is incorrect regarding the measured boiling point of this liquid?

- ☒ A- The measuring boiling point will stay the same because it's the same compound. ✗
- B- The boiling point will be lower at higher altitude due to the decreased atmospheric pressure.
- ☒ C- The boiling point will be higher at lower altitude due to the increased atmospheric pressure. ✓
- D- The boiling point will be higher at dead sea than Rum mountain. ✓

Q6: A sample of hydrate $\text{CoCl}_2 \cdot x\text{H}_2\text{O}$ is heated in a crucible. After heating, a blue powder of CoCl_2 is produced. If the mass percent of CoCl_2 in hydrate = 60%. Calculate the water of crystallization (x) of this salt? (Atomic weights: H=1, O=16, Cl=35.5, Co=59 g/mol)

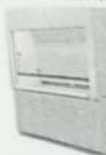
- A- 6
- ☒ B- 4
- C- 5
- D- 3

Q7: Calculate (the mass of water) in $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, if mass of anhydrous salt = 7.5g. (Molar mass of H_2O = 18, CaSO_4 = 136 g/mol).

- A- 2.26g
- B- 1.51g
- ☒ C- 1.99g
- D- 2.98g

Q8: Which one of the following is a physical property?

A-

**PART2:**

Answer the following questions

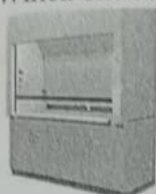
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Q8: Which one of these equipment used to extinguish or control small fire accident in lab?

A-



B-



C-

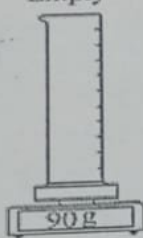
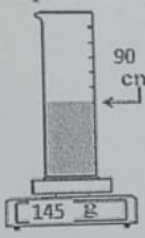
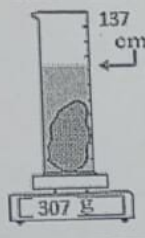


D-



PART2:

Answer the following question's and transfer your answer to the table in the cover page

Q		Answer
1	This Pictogram stands for 	oxi
2	According to the following picture, the <u>density of stone</u> equals: Empty  90 g Liquid added  90 cm³ Stone added  137 cm³	1.53 g/cm ³
3	What <u>physical observation</u> do you see when <u>I₂ is mixed with water</u> ?	
4	What is the way you use in your lab to <u>heat flammable liquids</u> ?	bath water or bath oil
5	The name of this equipment, is 	crucible
6	How is the value of X (moles of water) of a hydrated salt affected if water is completely <u>driven off during heating</u> ?	decrease
7	If a liquid sample contains some <u>glass impurities</u> , this cause the boiling point of liquid to: (increase, decrease, <u>not affected</u>)	

insoluble
21/2/19/25
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Good
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